Name: ___________________________

Homework Questions for Physics and Reality Week 2
Due the week of October 6
SHOW ALL WORK

Metric System Prefixes Table

T	"terra"	10^{12}
G	"giga"	10^{9}
M	"mega"	10^{6}
k	"kilo"	10^{3}
c	"centi"	10^{-2}
m	"milli"	10^{-3}
μ	"micro"	10^{-6}
n	"nano"	10^{-9}
p	"pico"	10^{-12}

Useful Values:

Gravitational constant, $G = 6.7 \times 10^{-11}\ m^3\ kg^{-1}\ s^{-2}$
Speed of light in a vacuum, $c = 3.0 \times 10^{8}$ m/s
Radius of the Sun, $R_S = 7.0 \times 10^{8}$ m
Mass of the Sun, $M_S = 2.0 \times 10^{30}$ kg
Mass of Jupiter, $M_J = 1.9 \times 10^{27}$ kg
Radius of Earth, $R_E = 6.4 \times 10^{6}$ m
Mass of Earth, $M_E = 6.0 \times 10^{24}$ kg
Radius of proton, $r_p = 8.4 \times 10^{-16}$ m
1 parsec (pc) = 3.26 light years (ly) = 3.09×10^{16} m

1. <u>Einstein's Theory of General Relativity</u>

1A. Which of the following statements describes why Einstein thought that Newton's Law of gravitation was not the best description of gravity? Select all that apply.

a. Newton's Law of gravitation was published in 1687, so it was outdated.
b. According to Newton, the force of gravity is instantaneous.
c. Newton did not provide a mechanism for how gravity exerts itself.
d. We can use Newton's Law to describe the motion of planets, rockets and projectiles to a high degree of accuracy.

1B. Which of the following situation pairs are consistent with the Equivalence Principle? Select all that apply.

a. A ball dropped in an accelerating rocket appears to fall to the "floor;" a ball dropped onto the Earth falls to the ground.
b. A train moving at a constant speed; a rocket at constant acceleration.
c. Light bends downward inside a closed laboratory on Earth; light bends downward in a closed rocket accelerating through space.
d. A person standing on Earth; a person standing on the surface of the Moon.

1C. (i) What observational evidence supports Einstein's General Theory of Relativity? Select all that apply.

a. The observation of gravitational waves by LIGO.
b. All observers in uniform motion can legitimately claim to be at rest.
c. The measurements of starlight deflection by the Sun during the Solar Eclipse in 1919.
d. The speed of light is constant.

1C. (ii) Describe in 2-3 sentences how the LIGO observatory detects gravitational waves.

2. <u>Black Holes and a Sense of Scale</u>

The radius of the event horizon of a black hole is called the Schwarzschild radius (r_s), and as you saw in lecture, it is given by the following equation:

$$r_s = \frac{2GM}{c^2}$$

where $\boldsymbol{M}$ is the mass of the black hole, $\boldsymbol{G}$ is the gravitational constant, and $\boldsymbol{c}$ is the speed of light.

Note that all the physical vales and constants you will need are given to you in the first page.

2A. (i) What would be the radius of a black hole with the same mass as Jupiter?

2A. (ii) Name an object or distance or place whose size is comparable to the size of the black hole you calculated above.

2B. Suppose that going through some telescope images, you discover a black hole with the radius of 15 cm. **Calculate the mass of this black hole. Which planet in our Solar System is closest in mass to this black hole?** Feel free to consult the internet.

3. <u>Gravity and the Warping of Time</u>

As you learned in lecture, time slows down near black holes. In fact, time slows down near all massive objects. Let's explore these effects.

After finally returning to Earth from the International Space Station, Sunita and Butch decide to take separate trips to explore the universe. Worried about loneliness, they decide to message each other every hour, according to their individual watches.

Butch and Sunita park their individual spaceships in separate empty corners of the universe.

3A. (i) According to Sunita how frequently is Butch messaging her?

a. Every hour
b. Less frequently than every hour.
c. More frequently than every hour.
d. It cannot be determined from the information provided.

3A. (ii) According to Butch how frequently is Sunita messaging him?

a. Every hour
b. Less frequently than every hour.
c. More frequently than every hour.
d. It cannot be determined from the information provided.

Butch continues to stay where he is, while Sunita ventures out near a black hole. The pair continues with their hourly messaging.

3B. (i) According to Sunita how frequently is Butch messaging her?

a. Every hour
b. Less frequently than every hour.
c. More frequently than every hour.
d. It cannot be determined from the information provided.

3B. (ii) According to Butch how frequently is Sunita messaging him?

a. Every hour
b. Less frequently than every hour.
c. More frequently than every hour.
d. It cannot be determined from the information provided.

4. <u>Gravitational Time Dilation</u>

Consider the gravitational time dilation formula, which relates the rate (***T***) at which time passes near any round massive object (not just a black hole) to the rate at which time passes in empty space (***t***):

$$T = t \times \left(1 - \frac{2GM}{c^2 r}\right)^{1/2}$$

where, once again, *G* is the gravitational constant, *M* is the mass of the massive object, and *r* is the distance to the center of the massive object.

4A. (i) While the gravitational time dilation formula is valid for any object with mass M, in the special case of Black Holes we can rewrite it in terms of their Schwarzschild radius r_s. Recall the formula for the Schwarzschild radius $r_s = \frac{2GM}{c^2}$. **Using this and the formula above, rewrite the gravitational time dilation equation for Black Holes in terms of r_s.**

4A. (ii) Using the rewritten equation from part 4.A, calculate the rate of passage of time at a distance equal to the Schwarzschild radius of a Black Hole ($r = r_s$), relative to the passage of time in empty space.

4B. Sunita parks her spaceship 5 Schwarzschild radii away from the center of the black hole at the center of our galaxy, while Butch continues to stay in empty space, very, very far away. Sunita conducts a spacewalk which takes her 250 minutes, based on her clock. **According to Butch, how many minutes was Sunita's spacewalk?** ***(****Hint: It may be best to use the equation from 4A. (i) for the special case of Black Holes.)*

5. <u>Galaxies and Hubble</u>

As you heard in lecture, Edwin Hubble discovered that the Universe is expanding. In Figure 1, we see the measurement of recession velocity (in units km/s) versus distance (in units of megaparsecs, abbreviated Mpc), where each circle represents the measurement for a single galaxy, derived from Type Ia Supernovae explosions. By measuring the slope of data similar to those shown here, Hubble was able to provide an estimate for the age of the universe.

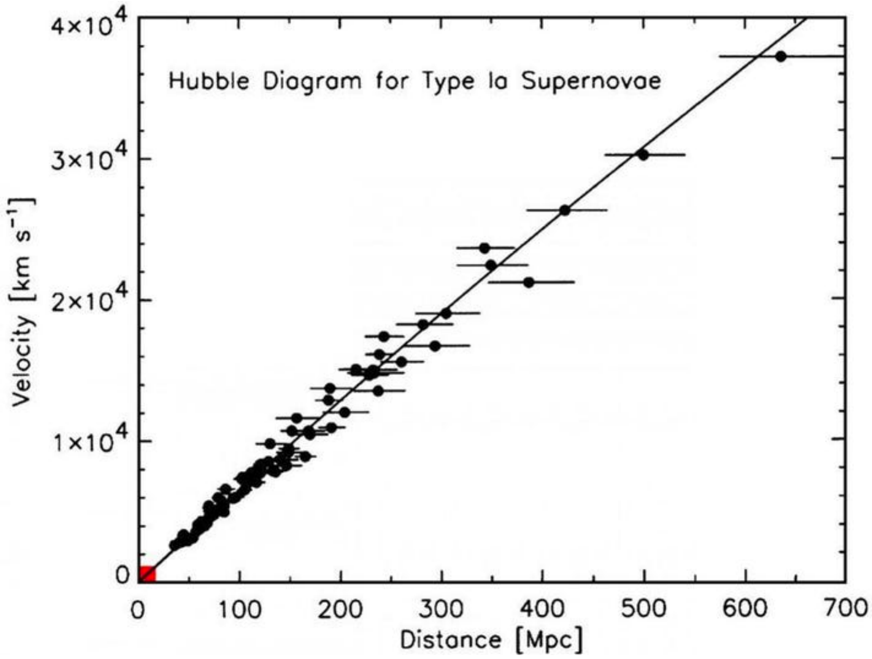

Figure 1. *Hubble Diagram for Type 1a Supernovae (Kirshner et al., 2004)*

5A. (i) In order for us to repeat Hubble's calculation, we need to first calculate the slope of the line in Figure 1. **Estimate the slope of the line and express it in units of $\frac{km}{s \cdot Mpc}$.**

5A. (ii) Convert the slope you found in part 5A. (i) to units of 1/s by converting the distance in Mpc to distance in km and canceling the units of distance.

5A. (iii) Lastly, the age of the Universe can now be estimated by finding the inverse of the quantity calculated in part 5A. (ii). **Using your previous answer, estimate the age of the Universe in Gyr (giga-years).**